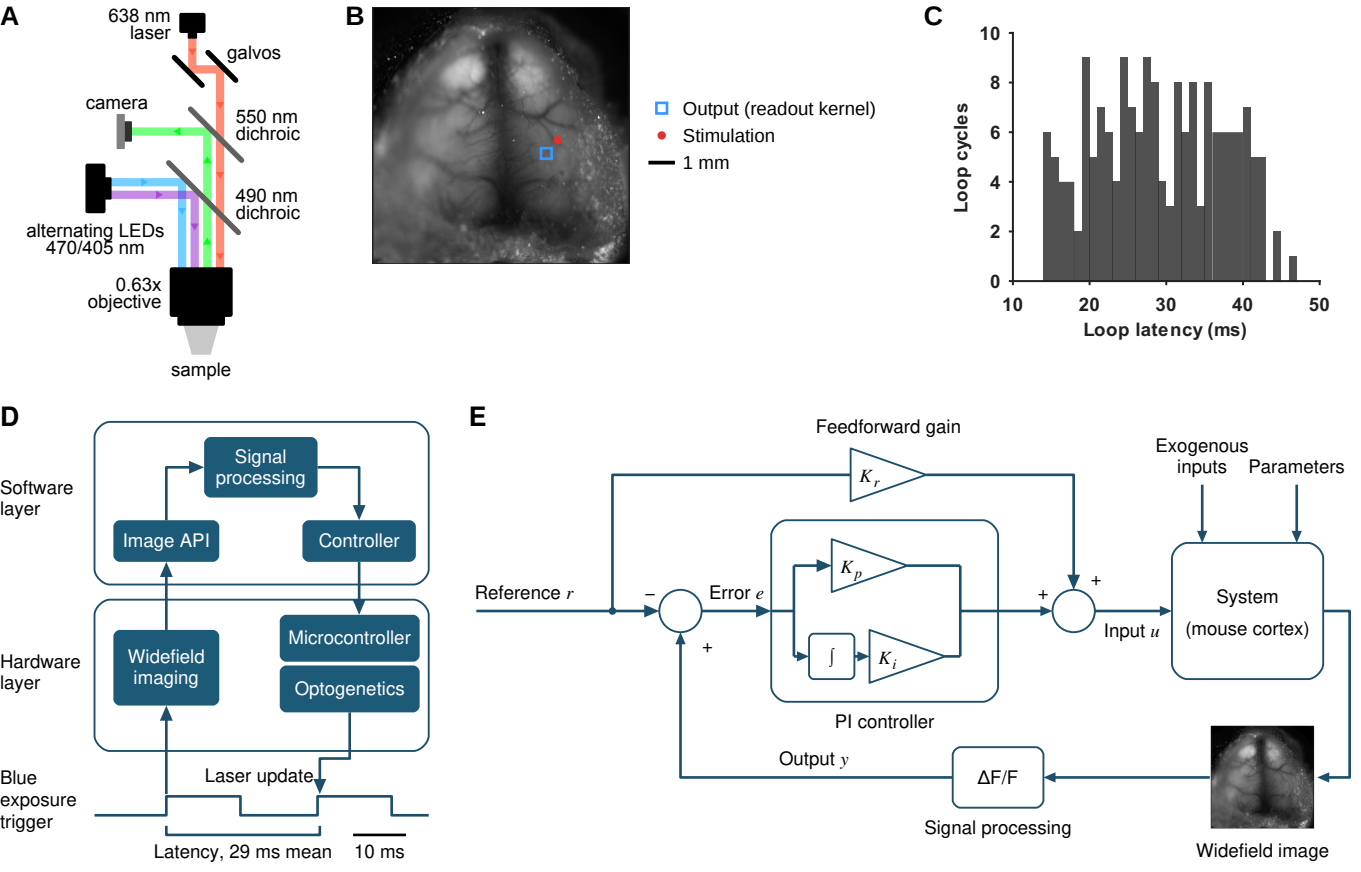

A
638 nm laser
galvos
camera
550 nm dichroic
490 nm dichroic
alternating LEDs 470/405 nm
0.63x objective
sample
B
Output (readout kernel)
Stimulation
1 mm
C
Loop cycles
Loop latency (ms)
D
Software layer
Hardware layer
Blue exposure trigger
Signal processing
Image API
Controller
Widefield imaging
Microcontroller
Optogenetics
Laser update
Latency, 29 ms mean
10 ms
E
Feedforward gain
Exogenous inputs
Parameters
Reference r
Error e
PI controller
Input u
System (mouse cortex)
Output y
ΔF/F
Signal processing
Widefield image